PORT-PROTECTION & NETWORK TIME PROTOCOLE

Note: By default, EC perform the Load-balancing according to the src-MAC address.

Sw1(config)#port-channel load-balance dst-mac --> *configure the EC to perform the Load-balancing according to the dst-MAC address.*

TIPs: On each of the endpoint of EC, we can define different load-balancing policy.

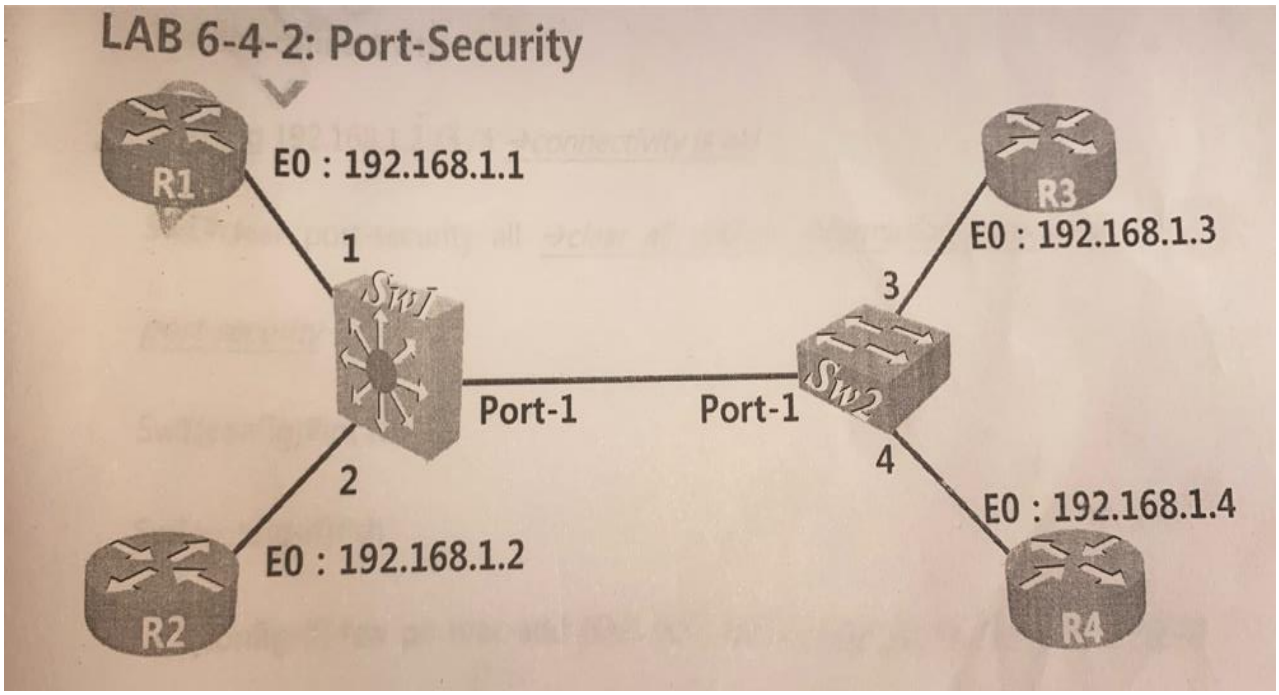
Port-Security

Port-security is a technology which can perform the Access-Control on the access-ports. By binding the legal MAC address to specific port (interface), illegal devices cannot access that port.

There are several limitations when we configure Port-security :

- Port-security cannot be configured on the Trunk port or the dynamic Access port.
- Port-security port cannot be the SPAN destination port;
- Port-security port cannot be the 802.1x port;
- Port-security port cannot be the member interface of any EtherChannel.

LAB 6-4-2: Port-Security



Step1: Basic IP configuration (Routers act as PCs in this LAB)

Omitted

Step2: On Sw1, enable port-security with Protect mode for Fa0/1, and use static configuration to bind the specific MAC address to the interface.

Sw1(config)#int fa0/1

Sw1(config-if)#sh

Sw1(config-if)#sw mo ac --> *Fa0/1 is a static Access interface*

Sw1(config-if)#switchport port-security --> *enable port-security*

Sw1(config-if)#switchport port-security mac-address 0000.0c4a.2946 --> *bind the legal MAC address*

Sw1(config-if)#switchport port-security violation protect --> *define the operation mode for port-security*

Sw1(config-if)#no sh

Test the connectivity:

R1#ping 192.168.1.2 /3 /4 --> *connectivity is ok!*

Sw1#clear port-security all --> *clear all cached information associated with port-security*

Sw1(config)#int fa0/1

Sw1(config-if)#sh

Sw1(config-if)#sw po mac-add 0001.0001.0001 --> *configure the new binding*

Relationship

Sw1(config-if)#no sh

Test the connectivity:

R1#ping 192.168.1.2 /3 /4 --> *connectivity is not Ok*

TIPs: The "Protect" mode of Port-security

Drop the frames which have the illegal source MAC address, but DO NOT shutdown the interface and DO NOT generate any information for the illegal frames.

TIPs: Fix the port-security configuration for Fa0/1.

Step 3: On Sw1, enable port-security with Restrict mode for Fa0/1, and use static configuration to bind the specific MAC address to the interface.

Bash

Sw1(config)#int fa0/2

Sw1(config-if)#sh

Sw1(config-if)#sw mo ac

Sw1(config-if)#sw po

Sw1(config-if)#sw po vi restrict # define the restrict manner for port-security

Sw1(config-if)#sw po mac-address 0000.0c75.e2e0

Sw1(config-if)#no sh

Test the connectivity:

R2#ping 192.168.1.1 /2 /3 --> *connectivity is ok*

Change the binding relationship

Bash

Sw1(config)#int fa0/2

Sw1(config-if)#sh

Sw1(config-if)#sw po mac-add 0002.0002.0002

Sw1(config-if)#no sh

R2#ping 192.168.1.1 --> *NOT OK*

Notice that there are the messages for the receiving of illegal frames.

TIPs: The "Restrict" mode of Port-security

Drop the frames which have the illegal source MAC address and generate any information for the illegal frames, but DO NOT shutdown the interface.

TIPs: check the status of Port-security

Sw1#show port-security interface fa0/x --> *check if the interface is enabled with port-security*

Sw1#show port-security address --> *observe the port-security MAC address*

Sw1#show int fa0/2 status --> *observe the status of specific interface*

TIPs: Fix the port-security configuration for Fa0/2.

Step 4: On Sw2, enable port-security with Shutdown mode for Fa0/3, and use static configuration to bind the specific MAC address to the interface.

Bash

Sw2(config)#int fa0/3

Sw2(config-if)#sw mo ac

Sw2(config-if)#sh

Sw2(config-if)#sw po

Sw2(config-if)#sw po vi shutdown --> *use "shutdown" mode for port-security*

Sw2(config-if)#sw po mac 0050.736c.0e3e

Sw2(config-if)#no sh

Test: R3#ping 192.168.1.1 /2 /4 --> *OK*

Bash

Sw2(config)#int fa0/3

Sw2(config-if)#sh

Sw2(config-if)#no sw po mac-add 0050.736c.0e3e

Sw2(config-if)#sw po mac-add 0003.0003.0003

Sw2(config-if)#no sh

Test: R3#ping 192.168.1.1 --> *NOT Ok*

Sw2/R3#show ip interface brief --> *the E0 of R3 is "status up, protocol down" ; the Fa0/3 of Sw2 is "status Down, protocol Down".*

Sw2#show int fa0/3 status --> *the accurate status of Fa0/3 is "err-disable"*

TIPs: The recovery of err-disable status

Bash

Sw2(config)#int fa0/3

Sw2(config-if)#sh # clear the err-disable status

Sw2(config-if)#no sh # re-enable the interface

Another method: auto-recovery

Sw2(config)#errdisable recovery cause psecure-violation --> *enable the auto-recovery for the reason of port-security violation.*

Sw2(config)#errdisable recovery interval 30 --> *define the auto-recovery interval to 30s (by default, 300s)*

TIPs :

Sw2#show err detect --> *observe which events will cause the err-disable status*

Sw2#show err recovery --> *observe the switch can auto-recover which type of err-disable*

TIPs: Fix the port-security configuration of Fa0/3 (omitted).

TIPs: The "Shutdown" mode of Port-security

If any illegal frame is received on the Port-security interface which is running with the "Shutdown" manner, the interface will be shut down immediately. Shutdown is the default manner of port-security interface.

Step 5: On Sw2, enable port-security with Protect mode for Fa0/4, and use Sticky method to bind the specific MAC address to the interface.

Bash

Sw2(config)#int fa0/4

Sw2(config-if)#sw mo ac

Sw2(config-if)#sh

Sw2(config-if)#sw po

Sw2(config-if)#sw po vi pro

Sw2(config-if)#sw po mac-add sticky

Sw2(config-if)#no sh

Note: Understand "Sticky"

R4#ping 192.168.1.1

Sw2#show run int fa0/4 --> *the sticky MAC address appear in the configuration file.*

Note: The differences between Sticky secure MAC address and Dynamic secure MAC address

Sticky MAC address will appear in the configuration file, if you save the configuration, even you reload the switch, the secure-MAC address will NOT change after the reloading; Dynamic secure-MAC address will NOT appear in the configuration file, after the reload, the switch need to learn secure-MAC address again.

NTP (Network Time Protocol)

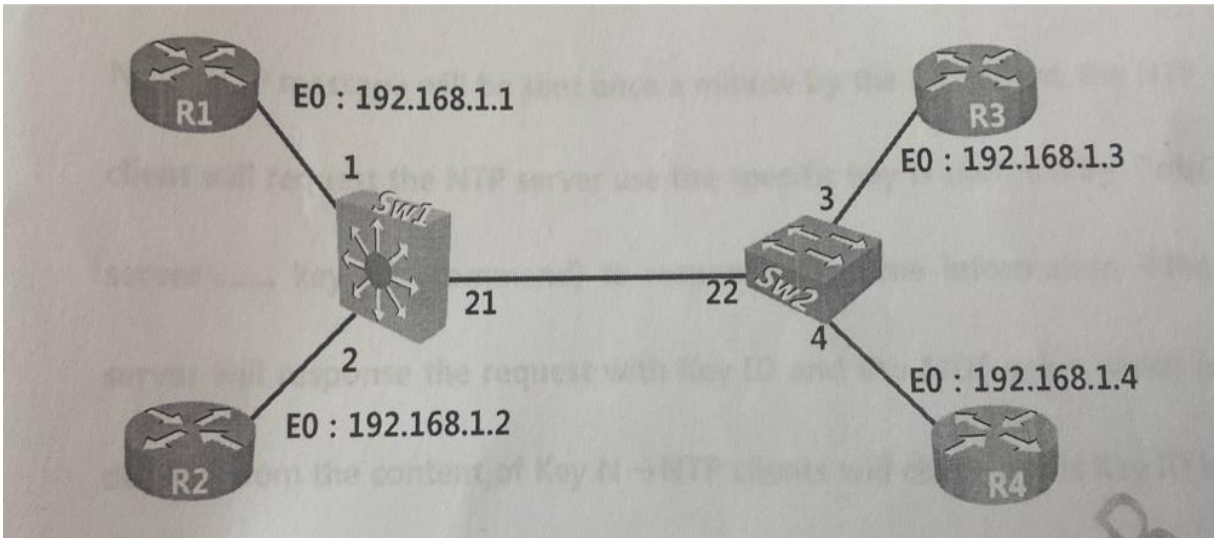
Time is IMPORTANT!

NTP is an Application-layer protocol which is supported by the port 123 of UDP. NTP can synchronize the time automatically and this is very helpful for the management of large scaled networks.

Important concepts:

- **Stratum Level:** Trustworthy of the time-source, lower value of the stratum level is more preferred.
- **NTP Server:** the entity which can provide time synchronization services to the NTP clients.
- **NTP Client:** the entity which request for the time synchronization from the NTP server.

LAB 6-4-3 : NTP(Network Time Protocol)



Step 1: Configure R1 to act as the time-source and NTP server

R1(config)#clock timezone GMT +8 --> *set the time-zone*

R1#clock set 16:53:00 5 Mar 2011 --> *set the local time*

TIPs : R1#show clock --> *observe the time information*

R1(config)#ntp master 3 --> *define the NTP stratum to "3"*

R1(config)#ntp authenticate --> *enable NTP authentication functionality*

R1(config)#ntp authentication-key 1 md5 cisco123 --> *define NTP key*

Step 2: Configure R2 to act as the NTP client, and synchronize with the specific NTP server

R2(config)#clock timezone GMT +8 --> *set the time-zone*

R2(config)#ntp server 192.168.1.1 key 1 --> *define the key which is for the communication with the NTP server*

R2(config)#ntp trusted-key 1 --> *define the trusted key (ONLY configure it on the NTP clients)*

R2(config)#ntp authenticate

R2(config)#ntp authentication-key 1 md5 cisco123

Note: NTP message will be sent once a minute by the NTP client, the NTP client will request the NTP server use the specific key N (defined by "ntp server ..... key N" command) to response the time information --> the server will response the request with Key ID and the MD5 value which is derived from the content of Key N --> NTP clients will check if this Key ID is trusted, if the key is trusted, NTP client will calculate the MD5 value from the local content of the specific key --> NTP client compare the received MD5 value and the local calculated MD5 value, if they are the same, NTP authentication is successful; if they are NOT same, NTP authentication is failed.

R2#show ntp status --> *Observe the NTP status*

R2#show ntp associations detail --> *observe the detailed information about the NTP associations*

TIPs: Observe the NTP activities

R2#debug ntp packets --> *Observe the sending and receiving of NTP packets*

R2#debug ntp authentication --> *Observe the NTP authentication*